

~~Do sample solution~~

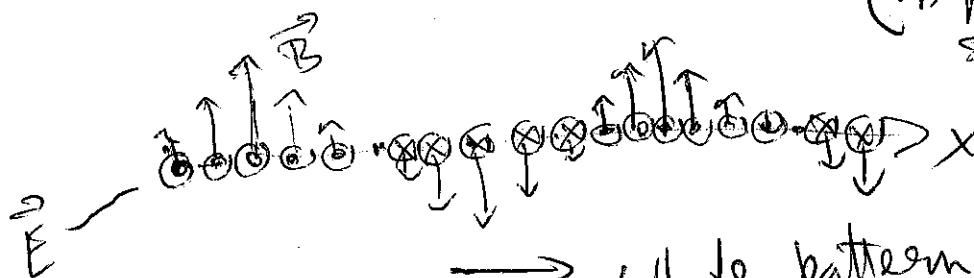
* An example solution:

$$\vec{E} = \hat{j} E_0 \sin(x-ct)$$

$$\vec{B} = \hat{k} B_0 \sin(x-ct)$$

~~Wave chosen frequency to be~~

(Frequency/wavelength specially chosen)



→ whole pattern moves forward with speed c .

~~The~~ The form above is a solution of

Maxwell's eqs: $\vec{\nabla} \cdot \vec{E} = \vec{\nabla} \cdot \vec{B} = 0$

$$\vec{\nabla} \times \vec{E} = \hat{k} \frac{\partial E_y}{\partial x} = \hat{k} E_0 \cos(x-ct)$$

$$\frac{\partial \vec{B}}{\partial t} = -\hat{k} c B_0 \cos(x-ct)$$

~~only~~ only if $E_0 = c B_0$.

$$\vec{E} = \hat{j} c B_0 \sin(x-ct)$$

$$\vec{B} = \hat{k} B_0 \sin(x-ct)$$

(74) In our solution,
 $\vec{E} \perp \vec{B}$, both are perpendicular to
direction of propagation.

$\Rightarrow$ general ~~features~~ features of
traveling E.M. waves (can be shown).

* Field pattern travels with speed c .

c = speed of light

$\Rightarrow$ Light is an electromagnetic wave.

~~The E.M. wave is a transverse wave.~~

The E.M. wave has a FREQUENCY f
and a wavelength $\lambda = \frac{c}{f}$

Generalize our simple solution!

$$\begin{aligned}\vec{E} &= \hat{j} c B_0 \sin\left(\frac{2\pi}{\lambda} x - \frac{2\pi}{\lambda} ct\right) \\ &= \hat{j} c B_0 \sin\left(\frac{2\pi}{\lambda} (x - ct)\right) = \hat{j} B_0 \sin(kx - \omega t)\end{aligned}$$

$$\vec{B} = \hat{k} B_0 \sin\left(\frac{2\pi}{\lambda} (x - ct)\right) = \hat{k} B_0 \sin(kx - \omega t)$$

* Depending on frequency, an electromagnetic wave can be a

γ -ray, X-ray, ultraviolet waves, visible light, infrared waves, microwaves, radio waves

* Spatial form of a traveling wave does not have to be oscillatory. Example of localized traveling pulse:

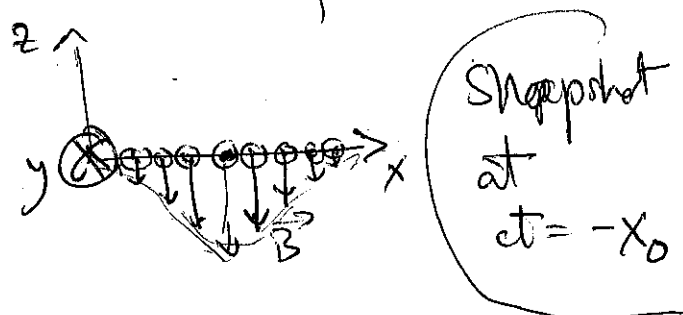
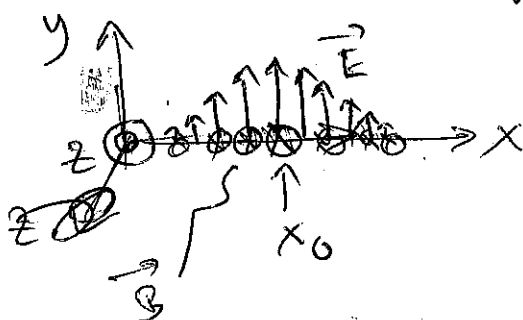
$$\vec{E} = E_0 e^{-\left(\frac{x+x_0}{\lambda_0}\right)^2} \hat{j}$$

$$\vec{B} = -\left(\frac{E_0}{c}\right) e^{-\left(\frac{x+x_0}{\lambda_0}\right)^2} \hat{k}$$

Traveling in -ve x direction.
Notice relative ~~minus~~ minus sign.

* Satisfies Maxwell's equations. (check)

Notice ~~used~~ used a Gaussian function.



Snapshot at $ct = -x_0$

Blob of $\vec{E}$ & $\vec{B}$ fields travels leftward with speed c .

* Another example: Feynman lectures II, Chs. 18 & 20

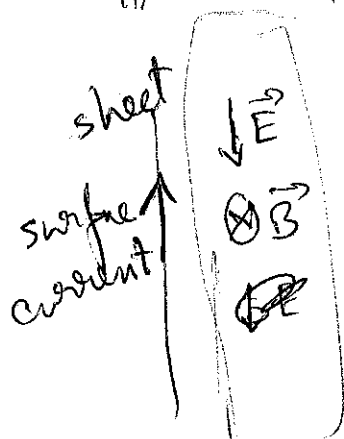


Sheet of current, turned rapidly on. ~~and off~~

Creates $\vec{B}$ -field, (increasing, larger nearer the sheet)

$\vec{B}$ -field creates $\vec{E}$ -field.

And so on...



$\vec{E}$ & $\vec{B}$ fields travel off.

→ ACCELERATING CHARGES CREATE E.M. radiation.

* STANDING WAVES

* Maxwell's Eqs. in empty space are LINEAR.

⇒ Sum of any two solutions is also a solution.

$$* \vec{E}_1 = \hat{j} E_0 \sin\left(\frac{2\pi}{\lambda}(x-ct)\right)$$

$$\vec{B}_1 = \hat{k} (E_0/c) \sin\left(\frac{2\pi}{\lambda}(x-ct)\right)$$

Travels

~~rightward~~
rightward

$$* \vec{E}_2 = \hat{j} E_0 \sin\left(\frac{2\pi}{\lambda}(x+ct)\right)$$

$$\vec{B}_2 = -\hat{k} (E_0/c) \sin\left(\frac{2\pi}{\lambda}(x+ct)\right)$$

Travels

~~leftward~~
leftward

The sum will also be a solution:

$$\vec{E} = \vec{E}_1 + \vec{E}_2, \quad \vec{B} = \vec{B}_1 + \vec{B}_2$$

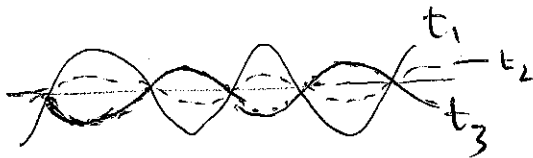
Use $\sin(\alpha + \beta) + \sin(\alpha - \beta) = 2\sin\alpha\cos\beta$

$$\sin(\alpha + \beta) - \sin(\alpha - \beta) = 2\cos\alpha\sin\beta$$

$$\Rightarrow \vec{E} = 2\hat{j} E_0 \sin\left(\frac{2\pi x}{\lambda}\right) \cos\left(\frac{2\pi c}{\lambda} t\right)$$

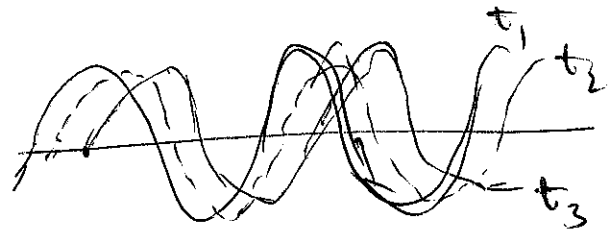
$$\vec{B} = -2\hat{k} \left(\frac{E_0}{c}\right) \sin\left(\frac{2\pi x}{\lambda}\right) \sin\left(\frac{2\pi c}{\lambda} t\right)$$

Standing



$$t_1 < t_2 < t_3$$

Traveling



$$t_1 < t_2 < t_3$$

~~EM WAVES~~

* MORE ON ^{EM} WAVES, p. 80a

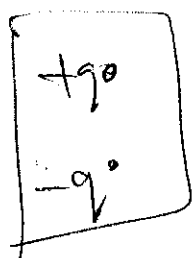
Electric Dipoles & Magnetic Dipoles

Electric "monopole" = Electric charge.

Electric dipole $\approx$ ~~two separated charges~~
configuration with separated positive & negative charges.

Simplest example: two point charges $+q$ & $-q$, separated by distance d .

Dipole moment: $|\vec{p}| = qd$, direction = -ve to +ve.



Field far away:

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \left(\frac{3(\vec{p} \cdot \vec{r})\vec{r}}{r^5} - \frac{\vec{p}}{r^3} \right)$$

On dipole axis: $\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{2\vec{p}}{r^3}$ {WORK OUT explicitly}

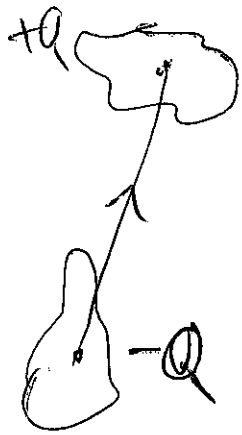
On perpendicular direction: ($\vec{p} \perp \vec{r}$):

$$\vec{E} = -\frac{1}{4\pi\epsilon_0} \frac{\vec{p}}{r^3}$$

At moderate distances, $\vec{E}$ will deviate

from far-field expression.

→ TORQUE on dipole due to $\vec{E}$ -field: $\tau = \vec{p} \times \vec{E}$



More involved ~~the~~ electric dipole.

$$\vec{p} = Qd$$

d = distance between "centres" of two charge ~~bits~~ clusters.

* Electric quadrupoles:

Quadrupole moment is a TENSOR

$$\begin{bmatrix} +q & 0 & -q \\ -q & 0 & +q \end{bmatrix}$$

OR

$$\begin{bmatrix} +q & 0 & -q \\ 0 & -q & 0 \\ +q & 0 & -q \\ 0 & 0 & -q \end{bmatrix}$$

~~Long-distance electric fields:~~ Long-distance electric fields:

$$E \propto \frac{1}{r^2} \text{ (monopole)}, \quad E \propto \frac{1}{r^3} \text{ (dipole)}$$

$$E \propto \frac{1}{r^4} \text{ (quadrupole)}, \dots \text{ etc.}$$

$$\rightarrow V(r) \propto \frac{1}{r} \text{ (monopole)} \dots$$

* Magnetic dipole

Expanding $V(r)$ in powers of r ! [MULTIPLY]
[EXPANSION]

$$V(r) = \frac{1}{r} + \frac{1}{r^2} + \frac{1}{r^3} + \dots$$

Object That creates far magnetic field:

$$\vec{B} = \frac{\mu_0}{4\pi} \left(\frac{3(\vec{m} \cdot \vec{r})\vec{r}}{r^5} - \frac{\vec{m}}{r^3} \right)$$

80

Similar profile of fields.

$\vec{B} \uparrow \rightarrow$

$\downarrow$

Simplest example:
current loop.

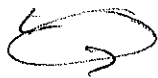
$\vec{m} \uparrow$

$\downarrow \vec{B}$

$|\vec{m}| = \text{current} \times \text{area of loop}$

$\uparrow \vec{B} \rightarrow \vec{B}$

$\downarrow \vec{B}$



$\downarrow \vec{B}$

* Reminiscent of field from a permanent magnet. ~~It~~ $\Rightarrow$ permanent magnets consist of current loops at atomic level.

End Interlude

E.M. waves continued

80a ~~80a~~

$$\begin{aligned}\vec{E} &= E_0 \sin(kx - \omega t) \hat{i} \\ \vec{B} &= (E_0/c) \sin(kx - \omega t) \hat{j}\end{aligned} \quad \left. \vphantom{\begin{aligned}\vec{E} &= E_0 \sin(kx - \omega t) \hat{i} \\ \vec{B} &= (E_0/c) \sin(kx - \omega t) \hat{j}\end{aligned}} \right\} \begin{array}{l} \text{wave propagating} \\ \text{in } \hat{k} \text{ direction} \\ (\text{direction of } \vec{E} \times \vec{B}) \end{array}$$

$k = \frac{2\pi}{\lambda}$ is the wavenumber

$\omega = 2\pi f$ is the angular frequency

$$k = \frac{2\pi}{\lambda} = \frac{\omega}{c}, \quad \lambda f = c \quad \left\{ \begin{array}{l} \lambda = \text{wavelength} \\ f = \text{frequency} \end{array} \right.$$
$$\frac{k}{\omega} = c$$

$$\vec{E} = E_0 \sin\left(\frac{\omega}{c}x - \omega t\right) \hat{i}$$

$$\vec{B} = \left(\frac{E_0}{c}\right) \sin\left(\frac{\omega}{c}x - \omega t\right) \hat{j}$$

* Sine or cosine? only difference in ~~phase~~

PHASE:

$$\sin\left(u + \frac{\pi}{2}\right) = \cos u$$

$$\sin\left(kx - \omega t + \frac{\pi}{2}\right) = \cos(kx - \omega t)$$

→ a shift of position/time.

~~806~~ 806

* $\vec{E}$, $\vec{B}$ and propagation direction are perpendicular.

→ EM waves are TRANSVERSE waves.

Contrast: sound waves are ~~longitudinal~~
LONGITUDINAL waves.

* If wave travels in x-direction,

$\vec{E}$ -field can be in either y- or z-direction → POLARIZATION.

If both transverse directions are ~~present~~
present, wave is said to be UNPOLARIZED.

Ex. Sunlight is unpolarized.

A "polarizer" ~~removes~~ cuts off one of the transverse directions.

ENERGY & ENERGY FLOW IN ELECTRODYNAMICS & MAGNETISM

(81)

Energy density due to $\vec{E}$ -field: $\frac{1}{2} \epsilon_0 E^2$

Due to $\vec{B}$ -field: $\frac{1}{2} \frac{1}{\mu_0} B^2$

$$u = \frac{1}{2} \epsilon_0 E^2 + \frac{1}{2} \frac{1}{\mu_0} B^2 = \frac{1}{2} (\epsilon_0 E^2 + \frac{1}{\mu_0} B^2)$$

(Often derived in the setting of CAPACITORS and INDUCTORS. We skipped these topics.)

Fields carry their own energy, even in free space! So EM waves should transport energy on its own.

Define Poynting vector:

$$\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B} = \epsilon_0 c^2 \vec{E} \times \vec{B}$$

$\Rightarrow$ Acts as a "current" of energy.

In free space, can derive from Maxwell's eqs.:

$$\boxed{\vec{\nabla} \cdot \vec{S} + \frac{\partial u}{\partial t} = 0}$$

(82)

→ Continuity equation for energy of EM field in free space

Derivation from Maxwell's equations. ~~also~~ →

~~problem set 11~~ problem set 11

* In the presence of currents (not in free space),

$$\vec{\nabla} \cdot \vec{S} + \frac{\partial u}{\partial t} + \vec{E} \cdot \vec{J} = 0$$

Use the identity $\vec{\nabla} \cdot (\vec{C} \times \vec{D}) = \vec{D} \cdot (\vec{\nabla} \times \vec{C}) - \vec{C} \cdot (\vec{\nabla} \times \vec{D})$

OPTIONAL

* Proof: $\vec{\nabla} \cdot \vec{S} = \frac{1}{\mu_0} \vec{\nabla} \cdot (\vec{E} \times \vec{B})$

$$= \frac{1}{\mu_0} \left[\vec{B} \cdot (\vec{\nabla} \times \vec{E}) - \vec{E} \cdot (\vec{\nabla} \times \vec{B}) \right]$$

$$= \frac{1}{\mu_0} \left[\vec{B} \cdot \left(-\frac{\partial \vec{B}}{\partial t} \right) - \vec{E} \cdot \left\{ \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right\} \right]$$

$$= -\frac{1}{\mu_0} \left(\vec{B} \cdot \frac{\partial \vec{B}}{\partial t} \right) - \vec{E} \cdot \vec{J} - \epsilon_0 \left(\vec{E} \cdot \frac{\partial \vec{E}}{\partial t} \right)$$

$$= -\frac{1}{\mu_0} \frac{1}{2} \frac{\partial}{\partial t} (B^2) - \frac{\epsilon_0}{2} \frac{\partial}{\partial t} (E^2) - \vec{E} \cdot \vec{J}$$

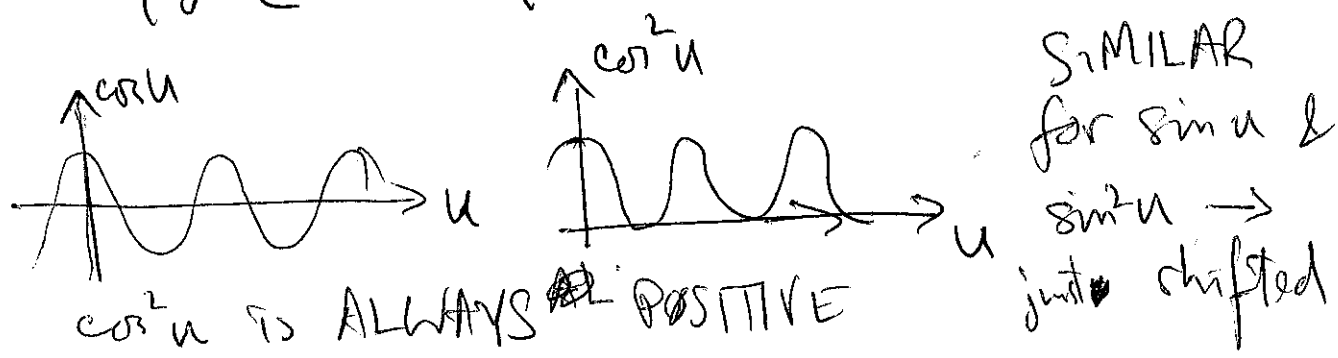
$$= -\frac{\partial u}{\partial t} - \vec{E} \cdot \vec{J}$$

* ENERGY in TRAVELING WAVES

$$\vec{E} = E_0 \cos\left(\frac{\omega}{c}x - \omega t\right) \hat{i}, \quad \vec{B} = \left(\frac{E_0}{c}\right) \cos\left(\frac{\omega}{c}x - \omega t\right) \hat{j}$$

Propagation in ~~the~~ $\hat{k}$ direction.

$$\vec{S} = \frac{1}{\mu_0} \frac{E_0^2}{c} \cos^2\left(\frac{\omega}{c}x - \omega t\right) \hat{k}$$



$\vec{S}$ is oscillatory but always points in direction of $\hat{k}$: direction of propagation of wave.

$\Rightarrow$ ^{Traveling} Waves carry energy in definite direction.

* ENERGY in STANDING WAVES

$\vec{S}$ oscillates direction. (cos or sin, not $\cos^2$ or $\sin^2$). Energy oscillates ~~not~~ locally but does not propagate.

84